

SATWIK VEMULA

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TECHNICAL SKILLS

Programming: Python, Java, Kotlin, C#, JavaScript/TypeScript, SQL

GenAI / LLMs: OpenAI GPT-4, Azure OpenAI, LangChain, LangGraph, Instructor, Prompt Engineering, ReACT Agents, Tool Calling, MCP

ML/AI Patterns: RAG, Agentic Workflows, LLM Classification, Decision Graphs, Semantic Layers, Evaluation Frameworks, AI-Augmented Development

Frameworks: FastAPI, LangChain, LangGraph, Pydantic, SQLAlchemy (async), .NET Core, Spring Boot, Cube.js

Orchestration: Camunda 8/Zeebe (BPMN), Event-Driven Architecture, Multi-Instance Parallel Processing

Data & Pipelines: BigQuery, Cosmos, PostgreSQL, Aerospike, Apache Spark, Kafka, Druid, Elasticsearch, Microsoft SQL, MySQL

Vector DBs: FAISS, Milvus, postgres

LLMOps / Observability: OpenTelemetry, Arize Phoenix, LangSmith, Prometheus, Splunk, Structured Logging, Inspect AI

Cloud & Infra: GCP (BigQuery, GCS, DataProc), AWS, Kubernetes, Docker, CI/CD

CRM / Enterprise: Salesforce (Sales Cloud, Apex, Workflows)

AI Dev Tools: Cursor, Claude Code, GitHub Copilot, Windsurf, Gemini, BMad

PROFESSIONAL SUMMARY

Staff Software Engineer with 10+ years of full-stack experience and 3+ years leading production GenAI/ML systems at enterprise scale. Architected LLM-powered agentic workflows, ReACT agents, and BPMN-orchestrated automation serving 1,500+ business users and unlocking \$500M+ in revenue impact (referenced in company earnings call). Built end-to-end AI services with custom evaluation frameworks, observability, and validation guardrails, while mentoring 8-10 engineers across cross-functional teams on GenAI patterns and AI-augmented development practices.

PROFESSIONAL EXPERIENCE

Walmart | Staff Software Engineer

January 2022 - Present

Wally - AI-Powered Enterprise Data Agent

Autonomous ReACT agent providing natural language data analysis to Sales, Merchandising, and Finance teams

- Architected a production ReACT agent (LangChain, OpenAI GPT-4, MCP) serving US and International business units across Sales, Merchandising, and Finance teams, reducing time-to-insight by 60%
- Built RAG pipelines using Milvus vector database and OpenAI embeddings for semantic search over store trait data and merchant knowledge base, enabling context-aware recommendations on assortment, pricing strategy, and store-level decision making
- Designed a Cube.js semantic layer abstracting 23+ BigQuery data models (financial, inventory, market share, pricing, store metrics) into an AI-friendly interface, eliminating manual SQL
- Built extensible skill/capability system (20+ tools: budget forecasting, supplier profiling, price gap analysis, store-DC alignment, product reviews) enabling rapid onboarding of new data domains
- Implemented MCP server integrations for cross-system data access, enabling the agent to retrieve and synthesize information from multiple enterprise sources in a single conversation
- Developed custom evaluation platform ("Proctor") with Inspect AI for automated LLM benchmarking: semantic query accuracy, measures recall, and canned response classification scoring
- Published shared walmart-wally library as internal PyPI package providing reusable ReACT architecture, LLM clients, and observability modules consumed by 4+ product teams

Assortment Gap Agent - Agentic Workflow Orchestration

Autonomous execution agent for merchandising workflow automation, cited in company earnings call

- Designed and led development of Camunda 8/Zeebe-orchestrated agentic system that processes millions of items across 6+ workflow types, contributing to \$500M+ in unlocked inventory value

- Built LLM-powered classification and enrichment pipelines converting 3P marketplace SKUs to 1P offers, directly increasing GMV
- Integrated with Salesforce (Sales Cloud) for vendor/partner master data (CMDA) during 3P-to-1P conversions, validating supplier eligibility, contract terms, and compliance status to automate outreach workflows
- Implemented fault-tolerant distributed workers handling orderability checks, replenishment activation, and offer metadata enrichment with configurable retry logic
- Resolved 2M+ transactability defects autonomously through rule-based decision engines combined with LLM validation guardrails
- Designed human-in-the-loop Action Center for merchant review of agent recommendations, balancing automation with oversight

Store Ticket Agent - Intelligent Ticket Processing & Resolution

End-to-end automated ticket resolution system processing 1M+ annual tickets from hundreds of US stores

- Built LLM-based ticket classification (LangChain, GPT-4) achieving 95%+ accuracy, auto-resolving 60% of tickets with LLM-generated merchant-facing explanations, and reducing average resolution time by 40%
- Designed LangGraph stateful decision graphs for automated root cause analysis across markdown, overstock, understock, and NOF ticket types
- Integrated with multiple enterprise CRM/ticketing platforms (OSC, ServiceNow, Salesforce Service Cloud) as unified intake sources, normalizing ticket data across channels via OAuth-authenticated REST APIs with retry logic and circuit-breaking patterns
- Pioneered spec-driven agentic development by integrating the BMad framework into the team's SDLC, using AI agents to generate and validate technical design documents, specifications, and test plans before implementation

Merchandising Analytics Platform (Item Table / Product Table)

Full-stack APIs and dashboards serving financial, inventory, and pricing data for 400M+ products

- Developed high-performance REST APIs serving financial KPIs (GMV, YOY performance, revenue opportunities) and designed Custom Rules engine (Kotlin) as a platform-level feature enabling complex query filters
- Migrated bulk data processing from Salesforce Apex batch jobs (eliminating governor limit violations and CPU exhaustion) to GCP Spark pipelines on DataProc, improving throughput and reducing platform resource contention

Charles Schwab | Software Engineer / Senior Specialist

August 2019 - January 2022

- Built the Income Reporting Service (.NET Core, C#) powering the primary portfolio landing page (web + mobile) for 30M+ trading customers, displaying real-time positions and performance
- Automated GCP DataProc Spark jobs processing 200M+ Kafka messages daily for income reporting pipelines
- Migrated services to support 8X load capacity during TD Ameritrade merger through horizontal scaling
- Implemented Elasticsearch queries over 1.5B+ documents with custom scoring logic for financial data retrieval

Granite Telecommunications | Software Engineer

November 2017 - August 2019

- Built EIS Order Portal for Federal office services from the ground up, securing \$5M+ in new contracts in year one
- Revamped Repair Management Portal reducing issue resolution time from 48 hours to 8 hours through online ticket intake automation

Bahwan Cybertek | Software Engineer

May 2016 - November 2017

- Modernized insurance product management from spreadsheets to web application, reducing product creation cycle time significantly

EDUCATION

Master of Science in Computer Science - University of Central Missouri, 2015